

Advanced Trigonometry



Trigonometric identities: Pythagorean and reciprocal

- 1 Given $\cos \theta = \frac{4}{5}$ and θ in the fourth quadrant, find $\sin \theta$ and $\tan \theta$.
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Sum and difference identities

- 2 Use the sum identity to find the exact value of $\sin(75^\circ)$, using $\sin(75^\circ) = \sin(45^\circ + 30^\circ)$.
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Double angle identities

- 3 Given $\sin \theta = \frac{3}{5}$ and θ in the first quadrant, find $\sin(2\theta)$ and $\cos(2\theta)$.
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Graphs and transformations of trigonometric functions

- 4 State the amplitude, period, and phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.
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Inverse trigonometric functions

- 5 Find the exact value of $\arcsin\left(\frac{1}{2}\right)$.
- 6 Find the exact value of $\arccos\left(-\frac{\sqrt{2}}{2}\right)$.
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Trigonometric equations with multiple solutions

- 7 Solve $2\cos^2 \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
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Applications: Law of Sines and Law of Cosines together

- 8 Two ships leave a port at the same time, one traveling at 20° and the other at 80° (measured from north), each at 15 mph. Find the distance between the ships after 2 hours, to the nearest mile, using the Law of Cosines.
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Polar coordinates

- 9 Convert the polar point $(4, \frac{\pi}{3})$ to rectangular coordinates.
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A multi-part trigonometric identity verification problem

- 10 This question has two parts. (a) Use the identity $\tan \theta = \frac{\sin \theta}{\cos \theta}$ together with the double angle formulas to derive a formula for $\tan(2\theta)$ in terms of $\tan \theta$. (b) Use your formula to find $\tan(2\theta)$ given $\tan \theta = 3$.
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Half-angle identities

- 11 Use the half-angle identity to find $\cos(15^\circ)$, given $\cos(30^\circ) = \frac{\sqrt{3}}{2}$.
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Solving trigonometric equations that factor

- 12 Solve $2\sin^2 \theta - \sin \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
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Area of a triangle using trigonometry

- 13 Find the area of a triangle with sides 8 and 12, and included angle 50° , using $\text{Area} = \frac{1}{2}ab\sin C$.
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Simplifying trigonometric expressions using identities

- 14 Simplify $\sin^2 \theta + \cos^2 \theta + \tan^2 \theta$ in terms of $\sec \theta$.
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A multi-part navigation and triangulation problem

- 15 This question has three parts. A hiker walks 5 miles due east, then turns and walks 7 miles at a bearing of 40° north of east. (a) Find the angle between the two legs of the hike (measured as the interior angle at the turning point). (b) Use the Law of Cosines to find the straight-line distance from the start to the end point, to the nearest tenth of a mile. (c) Explain why this distance is less than the total distance walked (12 miles).
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Verifying a trigonometric identity

16 Verify the identity $\frac{1 - \cos^2 \theta}{\sin \theta} = \sin \theta$.